Stats 506 F24 Midterm

Name: _____

UMID: _____

Question	Out of	Score
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____
5	_____	_____
6	_____	_____
Total	_____	_____

Instructions

Complete every problem. Use pen or pencil; closed notes.

Show intermediate work when possible. State any assumptions you make.

Minor syntax mistakes are acceptable when the intended calculation is clear.

Suggested practice time: 75 minutes, matching the F26 exam length.

Practice reconstruction from the official posted solutions. All answers and computed outputs have been removed. Original point allocations were not available in that source, so the score table is left blank.

https://dept.stat.lsa.umich.edu/~jerrick/courses/stat506_f24/midtermsolns.html

Problem 1

Determine q after each independent code fragment runs. Show intermediate work when useful. Assume required packages are loaded.

a.

```
x <- seq(2, 20, by = 2)
y <- x[x > 10]
q <- sum(y[y %% 4 == 0])
```

q = _____

b.

```
x <- list(a = 1:5, b = 6:10, c = 11:15)
y <- sapply(x, function(z) z[1] + z[length(z)])
q <- y * c(1, 10)
```

q = _____

c.

```
x <- matrix(c(2, 4, 6, 2, 4, 6), nrow = 2, byrow = FALSE)
q <- apply(x, 1, sum) - apply(x, 2, sum)
```

q = _____

Problem 1 (continued)

d.

```
q <- 0
x <- 4
while (x > 0) {
  for (y in 1:x) {
    q <- q + y
  }
  x <- x - 1
}
```

q = _____

e.

```
tibble(x = c(2, 4, 6, 8, 10),
       y = c(1, 2, 3, 2, 1)) %>%
  mutate(z = x * y, w = z - x) %>%
  group_by(y) %>%
  summarize(v = sum(w)) %>%
  ungroup() %>%
  arrange(desc(v)) %>%
  filter(v == 0) %>%
  select(y) %>%
  as.numeric -> q
```

q = _____

f.

```
f <- function(x, z = 0) {
  if (x <= 1) {
    return(z + x)
  } else {
    return(f(x - 1, z + x))
  }
}
q <- f(5)
```

q = _____

Problem 2

mydata is a data.frame with 5,732 rows and the following nine columns. Write R code for each cleaning operation, creating variables inside mydata and accounting for missing values.

Variable	Type	Example values
employeeid	int	1 2 3 4 5 6 7 8 9 10 ...
age	int	34 28 45 NA 38 30 31 27 42 36 ...
gender	chr	"Male" "Female" "Male" "" "Male" "Male" ...
educationlevel	chr	"Bachelor" "Master" "High School" "" "Bachelor" "PhD" ...
yearsatcompany	int	5 2 10 3 7 15 4 1 8 12 ...
department	chr	"Engineering" "HR" "Sales" "Marketing" "Engineering" ...
monthlyhours	int	160 150 170 155 165 175 160 145 168 172 ...
annualsalary	num	55000 48000 62000 50000 58000 67000 54000 ...
productivityscore	num	85 90 NA 88 80 -5 92 87 75 89 ...

a. Create a variable containing the log of annual salary.

b. Create a binary variable female. Gender is Male, Female, or unanswered; handle unanswered values appropriately.

c. Convert department to a factor.

Problem 2 (continued)

d. Calculate hourly wage from annual salary and monthly working hours.

e. Remove employees earning more than $50 per hour, using the variable from part d.

f. Standardize annual salary to mean 0 and variance 1.

g. Negative productivity scores identify exemptions. Recode those values as missing.

h. The minimum hiring age is 18. When the recorded age and years at the company imply an impossible hiring age, set both recorded variables to NA.

Problem 3

For a-d, write one R line fitting the indicated model. The data.frame is dat; all named columns are numeric.

a. $y = \beta_0 + \beta_1 x + \varepsilon$

b. $y = \beta_0 + \beta_1 q + \beta_2 z + \beta_3 (q \times z) + \varepsilon$

c. $y = \beta_1 q + \beta_2 (q \times z) + \varepsilon$

d. $y = \beta_0 + \beta_1 q + \beta_2 q^2 + \varepsilon$

Problem 3 (continued)

For e-g, write the mathematical regression equation represented by each call. Include an intercept when appropriate. z has numeric values 1, 2, and 3.

e.

```
lm(y ~ x + as.factor(z), data = dat)
```

f.

```
lm(y ~ I(x + 2) + I(z^2), data = dat)
```

g.

```
lm(y ~ x:z - x, data = dat)
```

Problem 4

Mark X when the pattern matches a date string; otherwise leave the cell empty. No string has leading or trailing whitespace. Patterns below are R string contents.

```
a. "^[^0-9]"
b. "(.)0.*\\1"
c. "^[0-9]{4}.?[0-9]{2}.?[0-9]{2}$"
d. "^[0-9]+.?[0-9]+.?[0-9]+$"
```

String	a	b	c	d
2023-10-14				
14/10/2023				
Oct 14, 2023				
10-14-2023				
2023/10/14				
14-Oct-23				
20231014				
2023.10.14				
14 October 2023				
14-10-2023				

Scratch work:

Problem 5

The tables data and data2 are shown below. For a-e, give each query output, including its column names.

data

a	b	c	i
5	4	1	1
8	4	1	2
4	4	2	3
9	4	1	4
3	8	2	5
1	5	1	6

data2

i	n
5	1
2	8
7	7

a.

```
SELECT *
FROM data
WHERE b == 8
```

b.

```
SELECT a, b - 1 AS b1
FROM data
ORDER BY -b1
LIMIT 2
```

Problem 5 (continued)

c.

```
SELECT c, SUM(a) AS aa
FROM data
GROUP BY c
HAVING aa < 15
```

d.

```
SELECT d1.a, d2.n
FROM data AS d1
INNER JOIN data2 AS d2 ON d1.i = d2.i
```

e.

```
SELECT d1.a, d2.n
FROM data AS d1
RIGHT JOIN data2 AS d2 ON d1.i = d2.i
```

Problem 5 (continued)

f. Write one SQL query producing the following result. Pay attention to which table supplies i.

a	i	n
9	4	NA
8	2	8
5	1	NA

Problem 6

A roulette wheel has 38 equally likely slots: numbers 1-36, 0, and 00. There are 18 red, 18 black, and 2 green slots. Represent 00 numerically as -1. In 1-10 and 19-28, odd numbers are red and even numbers are black; in 11-18 and 29-36 the parity colors are reversed. Both 0 and 00 are green.

You may use sample(x, size, replace, prob). It returns size draws from x; replace controls replacement and prob supplies optional relative weights. Input checks and function documentation are not required.

a. Write spin(n, color = FALSE). Return a vector of n spins: numeric slot values when color is FALSE, or character colors when TRUE. Assume n is a positive integer and color is logical.

Problem 6 (continued)

b. Use spin to simulate a fixed number of spins and count all length-three consecutive windows with the same color. Overlaps count: four consecutive reds contain two qualifying windows. Return the count, not a proportion. Your code need not be efficient; use of an efficient method can receive minor extra credit.